

16. Develop a C program for implementing random access file for processing the employee details

AIM:

To develop a C program for implementing random access file for processing the employee details

CODE:

```
#include <stdio.h>
```

```
#include <stdlib.h>
```

```
struct Employee
```

```
{
```

```
    int empId;
```

```
    char empName[50];
```

```
    float empSalary;
```

```
};
```

```
int main()
```

```
{
```

```
    FILE *filePtr;
```

```
    struct Employee emp;
```

```
    int choice;
```

```
    filePtr = fopen("employee.dat", "rb+");
```

```
    if (filePtr == NULL)
```

```
    {
```

```
        filePtr = fopen("employee.dat", "wb+");
```

```
if (filePtr == NULL)
{
    printf("Error creating the file.\n");
    return 1;
}

do
{
    printf("\nEmployee Database Menu:\n");
    printf("1. Add Employee\n");
    printf("2. Display Employee Details\n");
    printf("3. Update Employee Details\n");
    printf("4. Exit\n");
    printf("Enter your choice: ");
    scanf("%d", &choice);

    switch (choice)
    {
        case 1:
            printf("Enter Employee ID: ");
            scanf("%d", &emp.empId);

            printf("Enter Employee Name: ");
            scanf("%s", emp.empName);

            printf("Enter Employee Salary: ");
```

```
scanf("%f", &emp.empSalary);
```

```
    fseek(filePtr, (emp.empId - 1) * sizeof(struct Employee),  
SEEK_SET);
```

```
    fwrite(&emp, sizeof(struct Employee), 1, filePtr);
```

```
    printf("Employee details added successfully.\n");
```

```
    break;
```

```
case 2:
```

```
    printf("Enter Employee ID to display: ");
```

```
    scanf("%d", &emp.empId);
```

```
    fseek(filePtr, (emp.empId - 1) * sizeof(struct Employee),  
SEEK_SET);
```

```
    if (fread(&emp, sizeof(struct Employee), 1, filePtr) == 1)
```

```
    {
```

```
        printf("Employee ID: %d\n", emp.empId);
```

```
        printf("Employee Name: %s\n", emp.empName);
```

```
        printf("Employee Salary: %.2f\n", emp.empSalary);
```

```
    }
```

```
    else
```

```
    {
```

```
        printf("Employee not found.\n");
```

```
    }
```

```
    break;
```

case 3:

```
printf("Enter Employee ID to update: ");
```

```
scanf("%d", &emp.empId);
```

```
fseek(filePtr, (emp.empId - 1) * sizeof(struct Employee),  
SEEK_SET);
```

```
if (fread(&emp, sizeof(struct Employee), 1, filePtr) == 1)
```

```
{
```

```
printf("Enter Employee Name: ");
```

```
scanf("%s", emp.empName);
```

```
printf("Enter Employee Salary: ");
```

```
scanf("%f", &emp.empSalary);
```

```
fseek(filePtr, (emp.empId - 1) * sizeof(struct Employee),  
SEEK_SET);
```

```
fwrite(&emp, sizeof(struct Employee), 1, filePtr);
```

```
printf("Employee details updated successfully.\n");
```

```
}
```

```
else
```

```
{
```

```
printf("Employee not found.\n");
```

```
}
```

```
break;
```

case 4:

```

        printf("Exiting...\n");
        break;

    default:
        printf("Invalid choice. Please try again.\n");
    }

} while (choice != 4);

fclose(filePtr);

return 0;
}

```

OUTPUT:

```

Employee Database Menu:
1. Add Employee
2. Display Employee Details
3. Update Employee Details
4. Exit
Enter your choice: 1
Enter Employee ID: 1
Enter Employee Name: Indhu
Enter Employee Salary: 25000
Employee details added successfully.

Employee Database Menu:
1. Add Employee
2. Display Employee Details
3. Update Employee Details
4. Exit
Enter your choice: 2
Enter Employee ID to display: 1
Employee ID: 1
Employee Name: Indhu
Employee Salary: 25000.00

```